

ECON 0100 | Homework C2 | Taxes

Due: TBD

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely, and show your work so you can later understand your thought process. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Q1 | The Butterbeer Tax

The Ministry of Magic was in some financial trouble after their considerable expenditures during the war. They decided to impose a tax of 2 galleons on the sale of butterbeer as a source of funding. Use a graph to evaluate the welfare effects the policy had on the market. The supply and demand curves for butterbeer can be represented by the following equations.

$$P_b = 20 - \frac{1}{2}Q_b$$

$$P_s = 2 + \frac{1}{2}Q_s$$

Use a graph and algebra to calculate equilibrium before and after the tax. Start with the pre-tax equilibrium. Then use the tax equation ($P_b = \tau + P_s$) to solve for equilibrium quantity and the two prices. Conclude with the remaining relevant components. *Note: I'm looking for a complete description of the impacts on the market.*